

POFADDER, South Africa

WMO: 684160

Lat: 29.1236S

Lon: 19.3889E

Elev: 3222

StdP: 13.06

Time Zone: 2.00 (E02)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.3	38.6	6.5	8.6	65.9	9.4	10.0	66.2	18.3	51.9	16.0	49.4	5.2	140	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.9	97.9	61.7	95.8	61.1	93.6	60.4	67.9	82.5	66.6	83.1	65.3	83.0	7.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.8	99.9	74.5	61.1	90.8	74.9	58.4	82.3	74.8	34.2	82.3	33.1	82.5	32.0	83.1	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	16.3	14.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			29.0	69.9	1.6	1.4	27.8	70.9	26.9	71.8	26.0	72.6	24.8	73.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.5	79.4	78.6	76.3	69.2	62.5	54.3	53.2	57.0	62.6	68.6	72.4	76.8
	DBStd	11.34	5.66	5.31	6.55	7.13	6.51	6.03	6.54	7.43	7.29	7.99	7.84	6.09
	HDD50	92	0	0	0	0	3	24	44	19	2	0	0	0
	HDD65	1351	0	0	7	41	126	323	369	268	132	57	27	1
	CDD50	6484	913	801	816	577	389	153	142	235	380	577	671	830
	CDD65	2272	448	382	358	168	47	2	3	19	61	169	248	367
	CDH74	25088	5509	4382	3658	1271	329	16	19	191	598	1771	2811	4533
	CDH80	11618	2939	2191	1616	382	45	0	0	38	168	711	1260	2268
Wind	WSAvg	7.0	8.3	7.4	6.5	6.1	5.4	5.9	5.8	6.2	6.9	8.1	8.6	8.5
Precipitation	PrecAvg	4.4	0.5	0.8	0.7	0.7	0.3	0.2	0.2	0.1	0.1	0.2	0.3	0.3
	PrecMax	9.3	1.7	4.8	2.3	6.1	1.7	1.2	2.2	0.8	1.1	1.4	2.4	2.1
	PrecMin	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.2	0.6	1.0	0.6	1.3	0.4	0.3	0.4	0.2	0.2	0.3	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	101.6	99.5	96.6	90.7	84.7	76.3	76.0	84.9	89.4	95.9	96.5	99.5
		MCWB	61.8	63.8	62.3	59.6	56.4	53.0	51.7	54.5	56.7	59.4	58.7	60.9
	2%	DB	98.4	96.3	93.8	87.5	81.0	72.1	73.0	79.1	84.6	91.1	93.5	96.2
		MCWB	61.5	62.9	61.5	59.9	55.2	51.0	50.1	51.9	54.0	56.9	57.9	60.5
	5%	DB	95.9	94.0	91.4	84.4	77.9	69.0	69.5	74.9	80.9	87.4	90.5	93.6
		MCWB	61.1	62.2	60.8	58.4	54.1	49.5	48.6	50.4	52.8	55.8	57.2	59.8
	10%	DB	93.0	91.5	88.7	81.2	74.6	65.5	65.5	70.8	76.7	83.7	87.4	90.8
		MCWB	60.1	61.5	59.9	57.2	53.3	48.4	47.1	48.5	51.3	54.4	56.4	58.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.2	70.7	67.7	64.3	61.0	55.4	53.1	56.8	59.4	63.5	64.2	66.6
		MCDB	84.0	85.3	84.6	80.1	70.9	67.5	71.4	73.2	83.5	84.4	83.8	84.6
	2%	WB	67.9	68.3	66.1	62.5	58.2	53.1	51.2	54.0	55.9	60.2	61.5	65.4
		MCDB	81.5	83.0	84.9	78.0	73.3	66.5	69.8	74.7	79.7	81.4	82.2	84.2
	5%	WB	66.4	67.0	64.8	61.1	56.5	51.5	49.7	52.1	53.5	57.6	59.6	63.4
		MCDB	81.7	83.0	82.6	77.5	71.6	64.8	66.7	72.3	78.1	83.3	83.0	84.2
	10%	WB	64.5	65.5	63.4	59.7	54.6	49.8	48.1	49.7	51.8	55.6	57.9	61.5
		MCDB	83.6	83.0	82.0	76.3	72.6	62.9	63.0	68.3	75.6	81.4	83.6	84.3
Mean Daily Temperature Range	5% DB	MDBR	24.9	23.5	22.0	20.5	20.8	19.4	21.0	22.4	23.7	24.3	24.5	25.5
		MCDBR	26.4	25.2	22.4	21.7	23.1	22.7	24.3	25.7	26.6	26.6	26.4	26.9
		MCWBR	9.2	9.2	8.2	8.2	10.0	10.2	11.3	11.5	11.2	9.3	9.5	9.1
	5% WB	MCDBR	20.7	19.9	18.1	18.8	19.0	19.6	22.1	23.6	24.9	24.5	23.4	21.9
MCWBR		8.3	8.9	7.9	7.7	8.9	9.2	10.5	10.5	10.7	8.8	10.3	8.9	
Clear-Sky Solar Irradiance	taub	0.312	0.324	0.314	0.294	0.269	0.257	0.260	0.288	0.308	0.320	0.295	0.299	
	taud	2.496	2.474	2.488	2.539	2.588	2.583	2.561	2.438	2.359	2.359	2.503	2.514	
	Ebn at Noon	326	317	310	302	296	293	297	299	307	314	330	332	
	Edh at Noon	36	36	34	29	25	23	25	31	38	40	36	36	
All-Sky Solar Radiation	RadAvg	2715	2451	2106	1666	1320	1155	1248	1547	1960	2363	2686	2827	
	RadStd	100	93	66	48	45	42	36	51	68	61	81	93	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page